

Early Popularity Signals vs. Long Term Movie Success: A Comparative Analysis of Production and Audience Features

Prajwal Tejamurthy (A00050899)
Msc in Computing (Data Analytics)
Dublin City University
Dublin, Ireland
prajwal.tejamurthy2@mail.dcu.ie

Khushi Natraj Bijkal (A00049400)
Msc in Computing (Data Analytics)
Dublin City University
Dublin, Ireland
khushi.natrajbikal2@mail.dcu.ie

Aishwarya Shekar (A00050707)
Msc in Computing (Artificial Intelligence)
Dublin City University
Dublin, Ireland
aishwaryashekar.shekar2@mail.dcu.ie

Ashish Nagendrakumar (A00052416)
Msc in Computing (Artificial Intelligence)
Dublin City University
Dublin, Ireland
ashish.nagendrakumar2@mail.dcu.ie

Abstract—Predicting long-term movie popularity is a complex challenge due to the diversity of factors involved, from production characteristics to post-release audience behavior. This study investigates whether early audience engagement signals outperform traditional production-based features in predicting a movie's long-term success, measured by IMDb vote count. A dataset of 36,188 movies was constructed by integrating three publicly available sources: IMDb, MovieLens, and TMDB. Following the CRISP-DM framework, data preprocessing included removal of invalid records, outlier handling and log transformation of skewed variables. To address the research objective, features were divided into three groups: production-only, audience-only and a combined feature set. Two modelling approaches were applied - Linear Regression as a baseline model and Random Forest Regression to capture non-linear relationships. A comparative analysis was conducted across the three feature groups. Adding production features on top of audience features gave us a small but meaningful improvement from $R^2=0.80$ (audience only) to $R^2=0.863$ (combined). This suggests production context does add some predictive value, but the dominant signal still comes from audience engagement. Feature importance analysis revealed that MovieLens rating count is the dominant predictor, contributing approximately 82% of the model's predictive power. These findings demonstrate that early audience engagement is a stronger predictor of long-term movie success than traditional production features and that combining both feature types yields the most accurate predictions. The results have practical implications for streaming platforms, film investors and marketing strategists.

Keywords—Data Mining, Predictive Modelling, Movie Success Prediction, Regression Analysis, Feature Engineering, Data Integration, IMDb, MovieLens, TMDB.

I. INTRODUCTION

The film industry generates hundreds of billions of revenues annually, yet the commercial outcome of any individual movie remains highly uncertain [1]. Production decisions involving budget allocation, genre selection, cast composition, and release timing all carry significant financial risk, making early and reliable prediction tools extremely valuable to investors, studios and distributors.

Traditional approaches to predicting movie success have relied on historical trends, expert judgement, or simple statistical models based on production attributes [2]. While such methods provide a useful baseline, they fail to account for audience response and engagement, which are increasingly recognised as powerful indicators of future popularity [3].

Recent work has shown that machine learning techniques can leverage large, heterogeneous datasets to uncover predictive patterns that simpler models cannot capture [6]. In particular studies have demonstrated that early audience signals such as user ratings, review counts and platform activity are strongly correlated with long-term success [5].

This paper addresses the following research question: Do early audience engagement signals provide stronger predictive power than traditional production-based features for long-term movie popularity?

To answer this, we integrate data from three publicly available sources - IMDb, MovieLens, and TMDB and constructed a dataset of 36,188 movies spanning multiple decades, post performing EDA. We apply Linear Regression and Random Forest models across the three feature configurations: production, audience and combined. While prior studies highlight the importance of both production and audience-based features, they often evaluate them in isolation or on limited datasets. In particular, there is insufficient analysis on how early-stage audience signals, such as vote counts and ratings, compare directly with production attributes like budget and genre within the same modelling framework.

Additionally, the combined effect of these feature groups on predictive performance is not consistently examined across large, integrated datasets. This study addresses this gap by explicitly comparing production, audience and hybrid feature sets using a unified pipeline and consistent evaluation metrics. This is particularly important in practical settings where early-stage indicators are limited and decisions must be made before full lifecycle data becomes available.

The remainder of this paper is structured as follows. Section II reviews related literature. Section III describes the dataset and methodology used. Section IV presents and discusses the results. Section V concludes the study and outlines directions for future work.

II. LITERATURE REVIEW

A. Movie Success Prediction Using Production Features

Early research on movie success prediction primarily focused on production-related attributes such as budget, genre,

release timing and cast. These features were commonly used in statistical and machine learning models to estimate box office performance. One of the foundational studies by Sharda and Delen [11] demonstrated that neural network models can outperform traditional regression approaches by capturing nonlinear relationships between production variables and movie success.

More recent work has expanded this approach by introducing structured feature groups. Lash and Zhao [2] proposed a framework based on “who”, “what” and “when” features, which represent cast and crew characteristics, content attributes and release timing respectively. Their findings show that combining these feature groups improves early prediction accuracy and provides a more comprehensive understanding of movie performance.

Despite these advancements, models based only on production features have limitations. They rely entirely on pre-release information and do not capture audience response or behavioural patterns after release. As a result, their predictive performance may be restricted, especially when audience perception plays a significant role in determining success.

B. Audience-Based Prediction Approaches

Later research shifted focus towards audience-driven features such as user ratings, review counts and online engagement metrics. These features capture how audiences interact with movies and provide valuable signals about popularity and acceptance.

A key contribution in this area is the work by Mestyán et al., [5] which demonstrated that activity on platforms such as Wikipedia, including page views and edits, can be used to predict movie success even before release. Their study showed that early engagement patterns are strongly correlated with future box office performance, highlighting the importance of behavioural data.

In addition to activity-based approaches, network-driven methods have also been explored. Research on user interaction networks suggests that early user behaviour plays a role in how content spreads in terms of popularity [7]-[10]. Clustering behaviour in user networks has been linked to future popularity trends, indicating that audience structure can influence prediction outcomes.

Although audience-based models often achieve higher predictive accuracy, they typically focus only on post-release or early engagement signals and may overlook production-related factors. As a result, these models do not fully capture the complete set of variables that influence movie success.

C. Machine Learning Approaches for Movie Prediction

Recent studies have increasingly applied machine learning techniques such as Linear Regression, Random Forest, Support Vector Machines, and neural networks to model complex relationships in movie data. These methods are capable of handling large datasets and capturing nonlinear interactions between features, making them suitable for predicting movie success.

Lee et al. [1] introduced the Cinema Ensemble Model (CEM), which combined multiple classifiers and theory-driven features derived from transmedia storytelling. Their ensemble approach outperformed prior studies and

highlighted that feature engineering, not just algorithm choice, is central to improving predictive accuracy.

Fariha et al. [8] applied machine learning algorithms including Random Forest and Support Vector Machines to predict IMDb ratings for Netflix content, reporting high classification accuracy. Similarly, Bhadrashetty and Patil [3] used a data mining pipeline on IMDb data to predict movie ratings, demonstrating practical outcomes from relatively simple feature sets. Dixit et al. [9] applied exploratory data analysis combined with regression and classification models, confirming that data quality and preprocessing significantly impact model reliability.

Although more advanced deep learning models have also been explored, including neural networks and hybrid architectures, their practical use is often limited by the need for large datasets and reduced interpretability. In contrast, traditional machine learning models provide a better balance between performance and interpretability, especially in applications where understanding feature importance is valuable.

Overall, machine learning approaches provide a flexible and effective framework for movie success prediction, particularly when combined with well-designed feature engineering strategies.

D. Research Gap

Despite significant progress in movie success prediction, several limitations remain in existing studies. Many approaches focus either on production-related features or on audience-based signals but rarely examine their combined impact in a structured way. For instance, feature-based models emphasise elements such as cast, genre and release timing, while audience-driven approaches rely on engagement signals such as ratings and online activity. This separation limits the ability to capture the full set of factors that influence movie performance.

In addition, several studies rely on a single dataset, which restricts feature diversity and may reduce the generalisability of the results. Research has also shown that early engagement signals and network behaviour can provide valuable insights into future popularity, but these signals are often not integrated with production-level features in a unified model.

Another limitation is the lack of consistent comparison between statistical and machine learning approaches within the same framework. While machine learning models have demonstrated strong predictive performance, their advantages are not always evaluated alongside simpler statistical models using the same dataset and feature sets.

To address these gaps, this study integrates multiple datasets, including IMDb, MovieLens and TMDB, to create a more comprehensive feature space. It systematically compares production-based, audience-based and hybrid feature models using both statistical and machine learning approaches. This allows for a clearer understanding of how different feature groups contribute to prediction performance.

To support this analysis, Table I presents a comparison of existing studies in terms of feature types, modelling approaches and key limitations. The comparison highlights that most prior work does not fully integrate production and audience features within a single predictive framework, which motivates the approach proposed in this study.

Table I summarizes the key related works, their feature types, modelling approaches and identified limitations.

TABLE I. Summary of Related Work

Ref	Authors	Features	Model
[5]	Mestyán et al.	Audience	Regression
[2]	Lash & Zhao	Production	ML ensemble
[1]	Lee et al.	Mixed	Ensemble (CEM)
[6]	Liebig & Rao	Audience	Network - based
[8]	Fariha et al.	Audience	RF, SVM
[3]	Bhadrashetty & Patil	Mixed	Data mining
[9]	Dixit et al.	Mixed	EDA + ML

III. DATA MINING METHODOLOGY

A. Data Sources

This study uses three publicly available data sets to capture both production-related and audience-driven characteristics of movies:

1. **IMDb Dataset:** This provides core movie metadata such as title, release year, runtime, genres, as well as user ratings and vote counts. The latter serves as the target variable, as it captures long-term audience reach and engagement.
2. **MovieLens Dataset:** Maintained by GroupLens Research, contains user rating data, which is used to capture audience behavior and engagement patterns.
3. **TMDB Dataset:** It provides production related information such as budget, popularity scores and release details.

The integration of these datasets enables the construction of a comprehensive feature set that combines production attributes with audience engagement metrics. This multi-source approach improves feature diversity and allows for a more robust analysis of the factors influencing movie success.

The dataset spans a wide range of genres, languages, and release years, and includes both well-known and obscure titles. Vote counts follow a highly right-skewed distribution, reflecting the long-tail nature of movie popularity, where a small number of films accumulate the vast majority of audience attention. This characteristic motivated the use of log transformation for the target variable and several predictors.

B. Data Integration

The datasets were integrated using IMDb identifiers (*tconst*) as the primary key to ensure consistency across sources. As the MovieLens dataset uses a different identifier format (*imdbId*), a transformation step was applied to align it with the IMDb *tconst* format before merging. The TMDB dataset was linked using the *imdb_id* field, enabling accurate mapping between production data and metadata.

During integration, duplicate records and unmatched entries were removed to maintain data quality. Only movies with valid mappings across all datasets were retained, ensuring consistency across both production and audience-related features.

The resulting dataset combines production attributes such as budget, runtime and release information with audience-driven features including ratings, vote counts and engagement metrics. This unified structure allows both pre-release production factors and post-release audience behaviour to be analysed within a single framework.

By integrating multiple data sources, the dataset provides a more complete representation of movie performance. It also supports effective feature engineering and ensures that all models are trained on consistent and comparable data.

Figure 1 illustrates the end-to-end data mining pipeline for movie success prediction, from data collection and preprocessing to model training and evaluation.

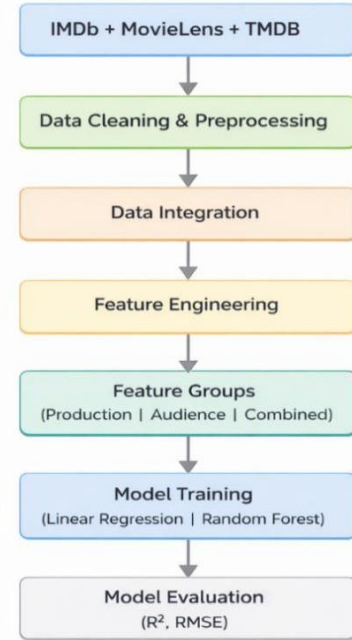


Fig 1. Data mining pipeline used for movie success prediction

C. Data Cleaning and Preprocessing

Data preprocessing was carried out to ensure data quality and consistency before modelling. The following steps were applied:

- Filtering the dataset to include only valid movie records.
- Removing entries with missing or invalid values in key attributes such as runtime and release year.
- Eliminating movies with zero votes to ensure meaningful target values.
- Removing outliers, including extreme runtime values and invalid release years.
- Handling missing values through imputation or removal, depending on the feature.

To further improve model stability, log transformations were applied to skewed variables such as vote count, budget, and rating count. This helped reduce the impact of extreme values and improved the distribution of the data.

In addition, unrealistic runtime values were filtered by restricting movies to a range of 40 to 240 minutes. Movies with future release years were also removed to maintain consistency in the dataset.

Movies with zero votes were excluded, as they offer no meaningful signal for the target variable. Outlier entries with future release years or extreme runtimes were filtered. Log transformation was applied to vote count, MovieLens rating count, budget and popularity score to reduce skewness and stabilize variance. Missing genre values were imputed with an 'Unknown' category.

D. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of key variables and identify patterns in the dataset. Several variables, including vote count and rating count, were found to be highly skewed, indicating that most movies receive low engagement while a small number achieve very high popularity.

As shown in Figure 2, the distribution of IMDb vote counts follows a right-skewed pattern.

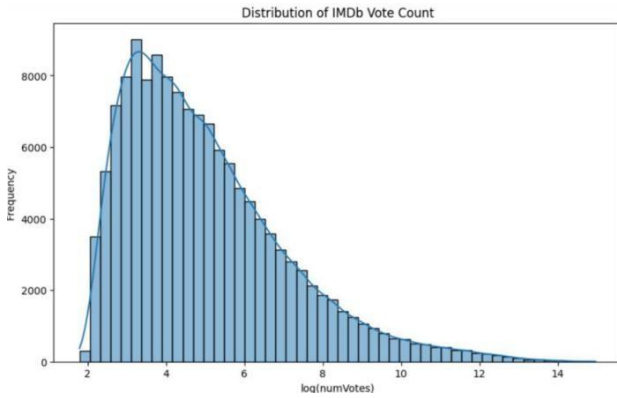


Fig 2. Popularity Distribution (IMDb Votes)

The majority of movies have relatively low vote counts, while only a few movies receive a significantly high number of votes. This imbalance highlights the presence of popularity concentration, where a small subset of movies dominates audience attention. To address this, log transformation was applied to reduce skewness and improve model performance.

In Figure 3, the distribution of IMDb ratings shows a near-normal distribution, with most movies receiving ratings between 5 and 7. This indicates that rating values are relatively stable compared to engagement metrics and are less skewed.

This skewed distribution suggests that raw vote counts alone may not be reliable indicators of popularity without appropriate transformation. Highly popular movies can disproportionately influence model training, leading to biased predictions if not addressed. By normalising these values, the model can better capture relative differences across the full range of movies rather than being dominated by a small number of extreme cases. This reinforces the need for balanced feature scaling to ensure that predictive models generalise effectively across both low and high-engagement movies.

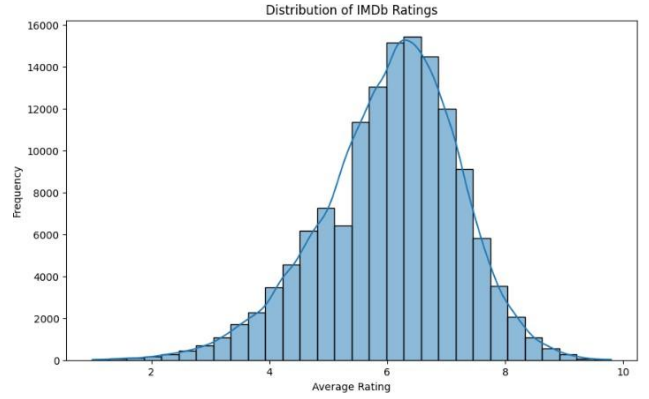


Fig 3. Distribution of IMDb Ratings

Following to that, the Figure 4 illustrates the relationship between early audience engagement and long-term success. A clear positive trend can be observed, where movies with higher early rating counts tend to achieve higher overall engagement. The density of points along the upward trend indicates a strong correlation between early user activity and long-term popularity.

These findings support the assumption that audience-driven features are strong predictors of movie success. The observed patterns justify the inclusion of engagement-based variables in the predictive models and highlight the importance of combining them with production-related features.

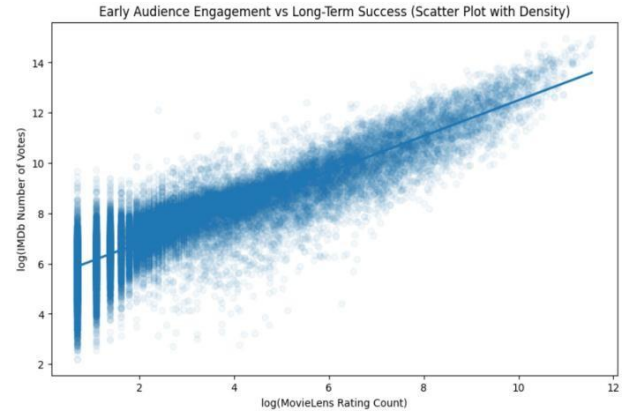


Fig 4. Early Engagement vs Long-Term Success

E. Feature Engineering

Feature engineering was performed to improve model performance and better represent both production-related and audience-driven aspects of movie success. Several new features were derived from the raw data to capture meaningful patterns:

- Log-transformed variables such as vote count, budget, and popularity to reduce skewness and stabilise variance
- Movie age, calculated as the difference between the release year and the current year, to capture temporal relevance
- Release month, to account for seasonal trends in movie performance
- Genre-based features, including genre count and primary genre extraction

- One-hot encoding of categorical variables such as genre and language to enable their use in machine learning models

support structured analysis, features were grouped into three categories:

1. **Production features:** budget, runtime, genre, release information
2. **Audience features:** ratings, vote counts and engagement metrics
3. **Combined feature set:** integration of both production and audience attribute. The target variable was defined as the logarithm of IMDb vote count, representing long-term movie success.

The target variable was defined as the logarithm of IMDb vote count, representing long-term movie success. This transformation helps reduce the impact of extreme values and ensures a more stable prediction target.

Overall, these feature engineering steps allow the model to capture both pre-release production factors and post-release audience behaviour, which aligns with recent research suggesting that combining these feature types improves prediction accuracy.

F. Modelling

Two modelling approaches were used to evaluate prediction performance: Linear Regression and Random Forest Regression. These models were selected to provide a balance between interpretability and predictive capability:

- **Linear Regression** was used as a baseline model due to its simplicity and interpretability. It allows for a clear understanding of the relationship between features and the target variable, making it suitable for analyzing the contribution of individual variables.
- **Random Forest Regression** was used to capture nonlinear relationships and complex feature interactions that cannot be modelled effectively using linear methods. As an ensemble-based approach, it improves prediction accuracy by combining multiple decision trees and reducing overfitting.

To enable a structured comparison, models were trained separately using three different feature sets:

- Production-only features
- Audience-only features
- Combined feature set

The dataset was split into training (80%) and testing (20%) sets to evaluate model performance on unseen data. This setup allows for a consistent and fair comparison across different modelling approaches and feature configurations. Model performance was evaluated using the coefficient of determination (R^2) and Root Mean Squared Error (RMSE). Feature importance was extracted from the best-performing Random Forest model to identify the most influential predictors.

Overall, the use of both linear and nonlinear models provides a comprehensive evaluation of the predictive power of different feature groups.

IV. IMPLEMENTATION

The implementation of the proposed system was carried out using Python, with a focus on building a clear and reproducible data mining pipeline. The workflow includes data loading, preprocessing, feature engineering, model training, and evaluation.

A. Tools and Libraries.

The following tools and libraries were used for implementation:

- **Python** as the primary programming language
- **Pandas** for data manipulation and integration
- **NumPy** for numerical operations
- **Matplotlib and Seaborn** for data visualisation
- **Scikit-learn** for model development and evaluation

These libraries provide efficient support for handling large datasets and implementing machine learning models.

B. Model Evaluation

The Model performance was evaluated using standard metrics such as R^2 and RMSE. These metrics provide insight into how well the models explain variance in the data and how accurately they predict movie success. The implementation ensures that all models are trained and evaluated under consistent conditions, allowing a fair comparison of different feature groups and modelling approaches.

C. Reproducibility

The entire workflow was implemented in a modular and structured manner, making it easy to reproduce results and extend the analysis. Each stage of the pipeline, from preprocessing to evaluation, was designed to be transparent and consistent.

This structured implementation supports reliable experimentation and ensures that the results can be validated and improved in future work.

V. EVALUATION AND RESULTS

The final dataset consisted of 36,188 movies containing both production-related and audience-driven features. Model performance was evaluated using both statistical and machine learning approaches to understand how different feature groups contribute to prediction accuracy.

Using linear regression, the combined feature model achieved the highest performance with an adjusted R^2 of approximately 0.47. The production-only model showed moderate performance with an adjusted R^2 of 0.37, while the audience-only model achieved an adjusted R^2 of 0.28. These results indicate that combining both feature types improves predictive capability compared to using either group independently.

To further evaluate model performance, Random Forest regression was applied. The combined feature model achieved the highest performance with an R^2 of approximately 0.86 and the lowest RMSE. This represents a significant improvement over linear models and suggests the presence of nonlinear relationships and feature interactions that are not captured by simpler models. Combining both sets pushed us to 86%. The extra 6% gain over audience-only shows that production features do add some value on top of audience signals, but they are not the main driver. The heavy lifting is still done by audience engagement.

Table II summarizes the performance of all 6 model configurations across the two algorithms and three feature sets.

TABLE II. Model Performance Summary

Sl no	Features	Model
1	Audience	Regression
2	Production	ML ensemble
3	Mixed	Ensemble (CEM)
4	Audience	Network - based
5	Audience	RF, SVM
6	Mixed	Data mining

Note: Best-performing model highlighted. Adj. R^2 reported for Linear Regression.

Across all configurations, Random Forest consistently outperformed Linear Regression, confirming the presence of non-linear relationships in the data. The production-only Random Forest achieved an R^2 of 0.29, indicating moderate predictive power from pre-release attributes alone. Budget and running time contributed most among production features, consistent with findings in prior work [2]-[4].

The audience-only Random Forest achieved an R^2 of 0.80, substantially exceeding the production-only model. This result supports the hypothesis that early engagement signals, particularly the volume of user ratings on MovieLens, are stronger indicators of long-term popularity than production characteristics. The combined Random Forest reached $R^2 = 0.86$ with the lowest RMSE of 0.70, representing the best overall model. Feature importance analysis identified MovieLens rating count as the dominant predictor, accounting for approximately 82% of total importance, followed by release year, IMDb average rating and runtime.

For Linear Regression, the audience-only model underperformed the production-only model suggesting that the relationship between early engagement and long-term success is inherently non-linear and not well captured by a linear model. The combined Linear Regression achieved the highest adjusted R^2 among the three linear configurations at 0.47, confirming that integrating both feature types improves performance even with simpler models.

These findings are consistent with Mestyán et al. [5], who showed that collective online activity is strongly predictive of commercial performance, and with Lee et al. [1] whose ensemble approach demonstrated the value of combining feature types. The dominance of the MovieLens rating count can be explained by its dual role: it reflects both the extent of early audience engagement and the breadth of a movie's pre-existing cultural footprint, making it a proxy for public awareness as well as audience reception.

VI. CONCLUSIONS AND FUTURE WORK

This study examined the prediction of movie success using both statistical and machine learning approaches based on production and audience-related features for long-term movie popularity using a dataset of 36,188 movies integrated from IMDb, MovieLens and TMDB. By integrating multiple datasets and applying a structured data mining process, the study provides a comprehensive view of the factors that influence movie performance. Both Linear Regression and Random Forest models were applied across three features

configurations, enabling a structured comparison of feature group contribution.

The results show that audience-related features, particularly early engagement signals such as rating counts, have a strong impact on prediction performance. Production features such as budget and runtime also contribute to the model, but their predictive power is lower when used independently. The combined feature model achieves the strongest performance, with an R^2 of 0.86 and an RMSE of 0.70 under Random Forest. MovieLens rating count was identified as the single most important feature at 82% of total importance, suggesting that early user engagement is a reliable and scalable signal for forecasting long-term success.

These findings have practical implications for film studios, streaming platforms, and marketing teams. Early monitoring of audience engagement following a movie's initial release or platform listing may enable more accurate forecasting and more targeted promotional investment.

However, the study has some limitations. Movie success is represented using vote count, which may not fully reflect financial outcomes such as box office revenue. In addition, important variables such as cast popularity, marketing expenditure and social media activity were not included due to data limitations.

Future work can extend this study by incorporating these additional features and exploring more advanced modelling techniques such as gradient boosting or XGBoost. Further validation strategies, including cross-validation and time-based evaluation, can also be used to improve model reliability. Extending the target variable to include streaming viewership data would also increase applicability to modern distribution contexts.

REFERENCES

- [1] K. Lee, J. Park, I. Kim, and Y. Choi, "Predicting movie success with machine learning techniques: Ways to improve accuracy," *Information Systems Frontiers*, vol. 20, pp. 577–588, 2018, [Available at](#)
- [2] M. T. Lash and K. Zhao, "Early predictions of movie success: The who, what, and when of profitability," *J. Manage. Inf. Syst.*, vol. 33, no. 3, pp. 874–903, 2016, [Available at](#)
- [3] A. Bhadrashetty and S. Patil, "Movie success and rating prediction using data mining," *J. Sci. Res. Technol. (JSRT)*, vol. 2, no. 1, pp. 1–4, Jan. 2024, [Available at](#)
- [4] V. R. Nithin, M. Pranav, P. B. Sarath Babu, and A. Lijiya, "Predicting movie success based on IMDb data," *Int. J. Bus. Intell. (IJBI)*, vol. 3, no. 2, pp. 34–36, 2014, [Available at](#)
- [5] M. Mestyán, T. Yasseri, and J. Kertész, "Early prediction of movie box office success based on Wikipedia activity big data," *PLOS ONE*, vol. 8, no. 8, p. e71226, 2013, [Available at](#)
- [6] J. Liebig and A. Rao, "Predicting item popularity: Analysing local clustering behaviour of users," *arXiv:1503.04404*, 2015, [Available at](#)
- [7] M. Agarwal, S. Venugopal, R. Kashyap, and R. Bharathi, "A comprehensive study on various statistical techniques for prediction of movie success," *arXiv:2112.00395*, Dec. 2021, [Available at](#)
- [8] F. Fariha, K. I. Rushee, and M. F. S. K. Arnop, "Prediction of IMDb rating in Netflix shows using machine learning algorithms," in *Proc. 3rd Int. Conf. Computing Advancements (ICCA)*, Dhaka, Bangladesh, Oct. 2024, pp. 754–761, [Available at](#)
- [9] P. Dixit, S. Hussain, and G. Singh, "Predicting the IMDB rating by using EDA and machine learning algorithms," *IJSRCSEIT*, vol. 6, no. 4, pp. 441–446, Jul.–Aug. 2020, [Available at](#)
- [10] C. D. Butler, M. Akhter, E. Jackson, M. A. Rahman, and L. Chang, "Predicting movie success using machine learning algorithms," in *Proc.*

15th LACCEI Int. Multi-Conf. Eng., Educ., Technol., Boca Raton, FL, USA, Jul. 2017 [Available at](#)

Networks Anal. Mining (ASONAM), Niagara, ON, Canada, Aug. 2013, pp. 1209–1214 [Available at](#)

- [11] R. Sharda and D. Delen, "Predicting box-office success of motion pictures with neural networks," *Expert Syst. Appl.*, vol. 30, no. 2, pp. 243–254, Feb. 2006 [Available at](#)
- [12] K. R. Apala, M. Jose, S. Motnam, C.-C. Chan, K. J. Liszka, and F. de Gregorio, "Prediction of movies box office performance using social media," in *Proc. 2013 IEEE/ACM Int. Conf. Advances Social*

Project link with code file & Datasets : Google Drive Link
